

MIMIR: Verifiable, Privacy-Preserving Exam Generation with Local Language Models

[Author 1], [Author 2], [Author 3]

[Affiliation to be completed]

[City], Hungary

[email addresses]

Abstract—The simplest way to obtain an exam from teaching material today is to upload the document to a chat-based large language model (LLM) and ask for a test. This gives no evidence that the keyed answers are supported by the document, no control over which parts of the material are covered, and it sends the material to a third party. We present MIMIR, an open system that generates multiple-choice exams from a teacher’s own documents with a small language model running on a consumer GPU. MIMIR replaces the single prompt by an explicit method: semantic chunking, a *blueprint* that assigns concepts, question types and Bloom levels to slots, per-question generation with chunk citations, and a verifier chain that checks grounding, distractors and blind answerability before a mandatory human review. Uploaded content is kept only for the duration of a job. We also contribute an evaluation methodology: ten experiment arms that switch components on and off, a 39-document Hungarian and English dataset with 326 reference questions, a cross-family LLM judge and document-level statistics. A pilot on one Hungarian document shows that planning improved grounding, blind answerability and distractor validity of the local 7B model and raised section coverage from 50 % to 58 %, that a large server model scored higher but drew its questions from only 25 % of the document’s sections, that the local model occupies 5.4–5.9GB of GPU memory, that a verifier run by the same 7B model does not yet separate good from bad questions, and that questions whose stem already contains the key inflate both automatic and LLM-judge scores. Run-to-run variation alone reaches 0.07 on the quality index, which is why the main study uses 39 documents.

Index Terms—automatic question generation, retrieval-augmented generation, verification, small language models, privacy, LLM-as-a-judge, educational technology, Hungarian

I. INTRODUCTION

Writing good multiple-choice questions is slow: the stem must be unambiguous, exactly one option must be correct according to the course material, and the distractors must be plausible but provably wrong. Large language models (LLMs) can draft such questions [1], [2], and the most common way teachers use them is direct: upload the lecture notes to a chat assistant and ask for “ten multiple-choice questions”. This is convenient, but it has four weaknesses that matter in a school.

(P1) Correctness is not checked. Nothing verifies that the keyed option is supported by the document or that the distractors are actually false; LLMs produce plausible but unsupported statements, which self-verification can reduce but not remove [3], [4]. **(P2) Structure and coverage are not controlled.** The requested number, types and difficulty of questions are only suggested in the prompt, and long inputs

are used unevenly—models attend to the beginning and end of long contexts more than to the middle [5]—so questions cluster on a few sections. **(P3) Privacy.** Teaching material may contain personal data, and a hosted assistant makes the provider a recipient of it. **(P4) Dependence on large hosted models.** Whole documents need long contexts and large models; the small models a school can run itself have limited context and memory.

MIMIR addresses these with a method rather than a prompt, and runs it on a single consumer GPU. We ask three research questions:

- RQ1** Does structured generation—planning, per-question grounding and verification—produce more valid and better-covering exams than prompting a model with the document?
- RQ2** What does each component contribute, and what does it cost in time and model calls?
- RQ3** How close does a 7B model on an 8 GB consumer GPU come to a large server-hosted model?

Contributions. (1) The MIMIR generation method: semantic chunking, a coverage-driven exam blueprint, cited per-question generation and a verifier chain (Section IV). (2) A privacy-preserving local deployment with zero retention of uploaded content and a measured GPU footprint that fits an 8 GB card (Section V). (3) An evaluation methodology and open harness: ten single-factor arms, a bilingual dataset with reference questions, a cross-family judge and document-level statistics (Section VI). (4) Pilot findings, including two measurement defects that would have invalidated earlier results and an estimate of run-to-run noise (Section VII).

II. RELATED WORK

Question and distractor generation. Surveys of educational question generation [1] and distractor generation [2] note that evaluation often relies on reference-overlap metrics or small human studies that rarely verify answer correctness. Small models can produce comparable educational questions after further pre-training and fine-tuning [6], [7], and small private models have been compared with large ones for Bloom-aligned assessment design [8]. We use off-the-shelf models without fine-tuning and add explicit grounding and verification.

Retrieval and long contexts. Retrieval-augmented generation (RAG) grounds output in retrieved passages [9]; hybrid

TABLE I
DESIGN GOALS: SENDING THE DOCUMENT TO A CHAT LLM VS. MIMIR.

Goal	Document → chat LLM	MIMIR
G1 Verifiable	Key not checked; no source reference	Every question cites chunks; grounding and blind-answer checks; unverified questions flagged
G2 Controlled	Count, types and difficulty only requested	Blueprint fixes count, types and Bloom levels; coverage-driven concept choice
G3 Private	Material sent to the provider	Local model; content deleted after each job; optional local-only mode
G4 Feasible	Long context, large model	Short calls within an 8,192–16,384-token context; 7B model on 8 GB
G5 Overseen	Output used as is	Mandatory review in an editor with the cited source; AI disclosure in the export

retrieval fuses BM25 [10] with dense embeddings [11] using reciprocal rank fusion [12], and graph-based RAG builds a concept graph for questions that need global context [13]. Long-context prompting, the alternative, uses the middle of the input poorly [5].

Verification and small models. Iterative self-feedback [3] and verification questions [4] reduce factual errors, and spending inference compute can substitute for model size [14]. Quantised 4-bit models lose little accuracy on general benchmarks [15], [16], which makes local deployment plausible.

LLM judges and languages. LLM judges correlate with humans [17] but favour their own family [18]. For Hungarian, benchmarks [19] and QA data [20] exist, but we know of no evaluation of exam generation.

III. DESIGN GOALS

Table I contrasts the direct approach with MIMIR on five goals derived from P1–P4 and from the human-oversight requirement of the EU AI Act [21].

IV. THE MIMIR METHOD

MIMIR is a set of services behind a gateway with a web frontend. Fig. 1 contrasts its generation pipeline with the naive RAG baseline, which retrieves the three chunks most similar to the request and asks for the whole exam in one call.

Document processing. Text is extracted from PDF, DOCX or plain text and split into semantic chunks: sentences are embedded with multilingual E5 [11], and a cut is placed where the cosine distance between neighbouring sentences exceeds a percentile (or $\text{mean} + k\sigma$) threshold, within a target size. The encoder works in windows of at most 512 tokens (Section VI-A). Chunks are indexed in a vector store that exists only for the current job.

Blueprint. One call proposes $n + 3$ concepts that cover the document, each with the chunk identifiers that support it. Slots are filled greedily, preferring concepts that touch chunks no slot covers yet, so the exam spreads over the document;

TABLE II
VERIFIER CHECKS, IN EXECUTION ORDER.

Check	LLM	Feedback on failure
Shape	–	MCQ: 4 options, 1 key
Meta-reference	–	Rephrase without “according to the text”
Citations	–	Cite a chunk from the given context
Grounding and distractors	1	Key not supported by the cited chunks, or a distractor may be true
Blind answer	1	A solver without the key chose another option
Leakage (new)	–	Key contained in the stem, or stem not a question
Duplicate	–	Too similar to an accepted question

the three spare concepts serve as replacements. Each slot receives a question type and a Bloom level [22] by largest-remainder allocation from a difficulty-dependent mix (easy: remember/understand; medium: understand/apply/analyse; hard: apply/analyse/evaluate). Count and types are therefore guaranteed by construction (G2).

Grounded generation. For each slot, the concept’s chunks are combined with the top- k retrieved passages (dense, or BM25 and dense fused with RRF), and the model writes *one* question that must cite the chunk identifiers supporting its key (G1). A concept-graph variant (E3) extracts concepts and typed relations per 3,000-character batch, replaces the planner call with graph concepts, offers sibling concepts as distractor ideas, and adds 1-hop neighbour chunks to the context.

Verification. Checks run cheapest first (Table II). A failed check returns its reason to the generator for one retry; after that the slot moves to a spare concept, and as a last resort the best attempt is kept and flagged as unverified for the teacher. Near-duplicates (embedding similarity > 0.9) are regenerated. After the pilot we added leakage checks, a repair call for malformed last-resort questions, and an optional verifier from another model family.

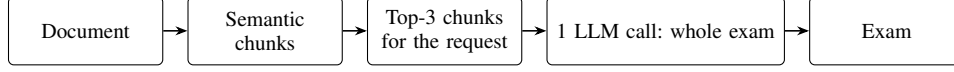
Human review. The draft opens in an editor where the teacher can edit, reorder, change the key, regenerate a single question and view each question’s cited source passage. Nothing is exported without approval, and the exported PDF states that the content was drafted with an AI system and by which model (G5).

V. PRIVACY AND LOCAL DEPLOYMENT

Zero retention (G3). Uploaded bytes, extracted text, chunks, embeddings and prompts exist only in service memory for the duration of a job; the vector store is cleared when the job ends or fails, and job results expire after 60 minutes. The audit log keeps the job identifier, model, prompt version, quality score and SHA-256 fingerprints and lengths of the request and context—never content. A `LOCAL_ONLY` switch disables every external model call; otherwise the upload screen names where the document will be processed and lets the teacher choose the local model.

Consumer GPU (G4). The generator is Qwen2.5-7B [23] with 4-bit (Q4_K_M) weights in the Ollama runtime, the only

E0 – naive RAG (baseline)



E1–E3 – MIMIR blueprint

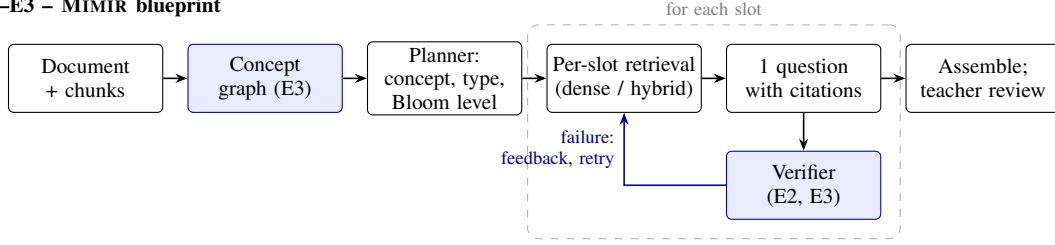


Fig. 1. Naive RAG versus the MIMIR blueprint pipeline. Blue: components switched on and off by the experiment arms. The blueprint fixes the exam structure before any question is written and verifies each question against its own cited context.

TABLE III

ESTIMATED GPU MEMORY AT AN 8,192-TOKEN CONTEXT (FP16 KEY-VALUE CACHE, ≈ 0.6 GB RUNTIME). WEIGHTS: GGUF FILE SIZE; CACHE: FROM THE PUBLISHED ARCHITECTURES.

Model	Quant.	Weights	Cache/token	Total	Card
Qwen2.5-3B	Q4_K_M	1.9 GB	36 KiB	2.8 GB	8 GB
Qwen2.5-7B	Q4_K_M	4.7 GB	56 KiB	5.8 GB	8 GB
Llama-3.1-8B [24]	Q4_K_M	4.9 GB	128 KiB	6.6 GB	8 GB
Qwen2.5-7B	Q8_0	8.1 GB	56 KiB	9.2 GB	12 GB
Qwen2.5-14B	Q4_K_M	9.0 GB	192 KiB	11.2 GB	12 GB

thing on the GPU; embeddings and all services run on the CPU. Because every call in the blueprint pipeline is short, the context stays at 8,192–16,384 tokens; the 7B model’s grouped-query attention needs 56 KiB of key–value cache per token, about 0.9 GB at 16,384 tokens. In the pilot, GPU memory rose from 1.6–1.9 GB (desktop, model not yet loaded) to a peak of 7.0–7.8 GB during generation, so the model and runtime occupied 5.4–5.9 GB—within the budget of an 8 GB card, including the desktop. Table III estimates the budget for other local models: 3–8B models at 4 bits fit 8 GB; a 4-bit 14B model or an 8-bit 7B model needs 12 GB.

VI. EVALUATION METHODOLOGY

The harness (`tests/eval`) runs in three steps: *run* generates exams for every document $\times$ seed of an arm, *score* computes metrics and calls the judge, and *report* produces tables, tests and figures. Each run stores its configuration, git commit and dataset version.

Dataset. Table IV summarises the 39 documents, frozen as `eval-v1`. EduQG [25] and MILQA [20] importers select short, medium and long documents and keep reference questions whose evidence sentence occurs in the document; references are used for retrieval recall and similarity, not as targets.

Arms. Each arm changes one factor (Table V). All local arms use Qwen2.5-7B (Q4_K_M, temperature 0.2); the server arms use a large Qwen-family model on the university GPU server.

TABLE IV

EVALUATION DATASET `eval-v1`.

Source	Lang	Docs	S/M/L	Median chars	Ref. Q
Own teaching texts	HU	3	3/0/0	1,923	12
OpenStax via EduQG	EN	12	4/4/4	16,752	74
Wikipedia via MILQA	HU	24	8/8/8	8,072	240
Total		39	15/12/12	7,919	326

TABLE V
EXPERIMENT ARMS.

Arm	Configuration
B-doc-L / S	Whole document in one prompt, local 7B / server model (status quo)
E0	Naive RAG: top-3 chunks, whole exam in one call (baseline)
E1	Blueprint: planner, per-slot retrieval, one question per call
E2	E1 + verifier chain with retries
E2x	E2 with a verifier from another family (Llama-3.1-8B)
E2f / E2h	E2 without blind answer test / with hybrid retrieval
E3	E2 + concept graph
E4 / E4b	E0 / E2 with the large server model
E5a / b / c	E0 with the legacy 512-token encoder / mean+1.5 σ cuts / fixed 800-char chunks

Measures. *Format compliance* is 1 if an exam has exactly the requested number and shape of questions. Per question, the judge decides *grounding* (the source supports the key, with a quoted sentence), *distractor validity* (each distractor false and on topic) and *blind answerability* (answering from the source without the key picks the keyed option), and gives 1–5 rubric scores. The *quality index* is the mean of these four rates. *Coverage* is the share of the document’s chunks that are the closest chunk (by embedding) of at least one question; we also record duplicates, meta-references, answer leakage, unverified questions, calls, tokens, time, peak GPU memory, the GPU model and the share of the model held in GPU memory.

Judge and teachers. The judge is gpt-oss-120b [26], from a different model family than the generators, to avoid self-

TABLE VI

PILOT: ONE HUNGARIAN DOCUMENT, 3 SEEDS, 30 QUESTIONS PER ARM.
RATES IN %.

Arm	Index	Form.	Grnd.	Blind	Dist.	Cov.	Leak.	Unv.	Calls	Min
E0	0.83	100	67	85	80	50	0	—	1	0.3
E1	0.88	100	73	93	87	58	0	2.0	23	3.0
E2	0.76	67	73	77	89	58	3	4.3	75	6.2
E2f	0.73	67	53	82	90	58	7	3.3	59	5.3
E2h	0.76	67	67	79	93	67	10	4.0	76	6.7
E3	0.85	67	87	100	87	50	20	6.0	101	8.0
E4	0.99	100	100	100	97	25	0	—	1	6.6
E5a	0.90	100	80	86	93	50	0	—	1	0.3
E5b	0.90	100	80	90	91	56	0	—	1	0.3
E5c	0.81	100	60	79	83	67	0	—	1	0.3

Form. format compliance; Grnd. grounding; Dist. distractor validity; Cov. coverage; Leak. key in stem; Unv. unverified questions per exam; Calls: mean model calls per exam; Min: median minutes per exam.

preference [18]. Blind teacher rating sheets use the same rubric; Krippendorff’s α [27] and Spearman’s ρ validate the judge.

Statistics. Documents are the unit of analysis: measures are averaged over questions and seeds per document, with 10,000-resample bootstrap CIs [28], paired Wilcoxon tests [29] with Holm correction [30], and rank-biserial effect sizes [31]. Fewer than six documents cannot reach $p < 0.05$, so single-document results are reported descriptively.

A. Measurement validity

Building the harness exposed two defects that had made earlier end-to-end measurements invalid. *Fallback exams counted as results*: when every model call failed, the backend returned a hard-coded one-question exam; a test run on 3 May 2026 reported all three documents as generated in 4–9 s, but each “exam” was this placeholder. It is now flagged and counted as a failure. *Truncated chunking*: the chunker embedded each document in one 512-token pass, so every sentence after roughly the first page received a zero vector. At 4.0–4.5 characters per token an estimated 68–71 % of the average eval-v1 document was affected. The encoder now works in windows; arm E5a keeps the old one to measure the effect. A third issue, answer leakage that inflates judge scores, was found in the pilot (Section VII).

VII. PILOT RESULTS

The pilot ran all local arms and E4 on one short Hungarian teaching text (1,702 characters, 4 chunks, easy difficulty) with three seeds and ten questions per exam (30 questions per arm), on a consumer GPU (model) in a desktop PC with a 16,384-token context. E4b did not complete. With one document no difference can be significant; the results show directions and costs, and they shaped the main study.

Run-to-run noise. E0 and E5a received identical chunks, identical retrieved context and the same seeds—the legacy and windowed encoders coincide on a document shorter than 512 tokens—yet produced different questions and indices of 0.83 and 0.90. GPU inference is not bit-reproducible even at low temperature, so on a single document differences below about

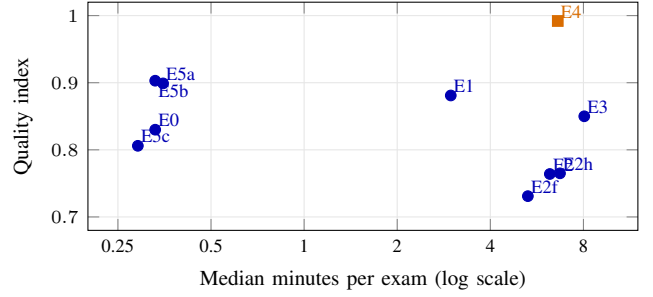


Fig. 2. Pilot quality versus time per exam. Blue: local 7B model on the consumer GPU; orange: large server model (naive RAG).

0.07 are noise. This is the main reason the study uses 39 documents and document-level statistics.

Planning helps (RQ1, RQ2). Compared with naive RAG, the blueprint (E1) improved grounding (67→73 %), blind answerability (85→93 %) and distractor validity (80→87 %) at unchanged format compliance, raised coverage from 50 to 58 % and the judge’s correctness score from 3.5 to 3.7, at 23 calls and 3.0 minutes per exam instead of one call and 20 seconds.

Verification with the same small model is not yet effective (RQ2). The verifier arms flagged 3.3–6.0 of 10 questions per exam as unverified. In one of the three exams of each of E2, E2f, E2h and E3, every attempt for a slot failed and the kept last-resort question had only two or three options; this set format compliance to zero for that exam and lowered its quality index by 0.25. Verification on top of planning did not raise grounding (73 % for E1 and E2) and lowered blind answerability (93→77 %). The concept-graph arm (E3) reached the highest local grounding (87 %), blind answerability (100 %) and judge correctness (4.4 of 5, against 3.5 for E0), but these scores are partly an artefact (see answer leakage below). Table VII shows why: pooled over the four verifier arms, the questions the 7B verifier passed were judged grounded *less* often (66 %) than those it flagged (75 %); they were blind-answerable somewhat more often (89 vs. 79 %), and the judge would use both groups equally often. Scored as a classifier of the judge’s grounding verdict, the verifier was at chance level in every arm (Cohen’s κ from -0.14 to 0.07 , recall 0.42 – 0.62). A verifier that is the generator itself is not independent of it [18], and a 7B model cannot yet decide reliably whether a passage entails a key. The pilot thus identified two fixes for the main study: the last-resort path must never emit a malformed question, and the verifier must be validated as a classifier—its agreement with the judge and the teachers is measured, and a stronger or different-family verifier is tested before its flag is shown to teachers.

Server versus local (RQ3). The large server model with naive RAG scored highest (0.99) but drew all its questions from a quarter of the document’s chunks (coverage 25 % vs. 50–67 % for local arms)—the clustering problem P2 at small scale. On the shared university server it needed a median of 6.6 minutes per exam, as long as the verified local pipeline

TABLE VII

DOES THE 7B VERIFIER AGREE WITH THE JUDGE? PILOT QUESTIONS OF E2, E2F, E2H AND E3, BY THE VERIFIER’S VERDICT; JUDGE OUTCOMES IN %.

Verifier verdict	<i>n</i>	Grounded	Blind correct	Would use
Passed (verified)	67	66	89	61
Flagged (unverified)	53	75	79	64

TABLE VIII

PILOT: JUDGE RUBRIC SCORES (1–5, MEANS OVER 30 QUESTIONS) AND SHARE OF QUESTIONS THE JUDGE WOULD USE IN AN EXAM.

Arm	Correct.	Clarity	Distract.	Bloom fit	Would use
E0 (naive)	3.5	4.7	3.5	4.8	63 %
E1 (blueprint)	3.7	4.0	4.0	5.0	67 %
E2 (+ verifier)	3.4	4.1	3.6	5.0	60 %
E3 (+ graph)	4.4	4.4	3.9	5.0	77 %
E4 (server, naive)	5.0	5.0	4.6	5.0	100 %
E5c (fixed chunks)	3.3	4.0	3.0	5.0	57 %

(6.2 minutes); the local naive baseline took 20 seconds.

The rubric scores (Table VIII) tell the same story in the teachers’ terms: the judge would use every question of the server model but 57–77 % of the local ones. Planning raised distractor and Bloom-fit scores at a small cost in clarity; one-question-at-a-time prompts produced longer stems (88 vs. 60 characters on average) and much longer options (315 vs. 60 characters for all four).

Answer leakage. Reading the questions (Table IX) exposed a failure that neither our metrics nor the judge penalised: some verified questions copy a source sentence into the stem and repeat it as the key. For keys of at least three words, the key’s words appeared in the stem ($\geq 80\%$ token overlap) in 20 % of E3’s questions, 7–10 % of E2f and E2h, 3 % of E2 and none of the single-call or E1 questions (Table VI, Leak.), and 47 % of E3’s stems were statements rather than questions (10–33 % in five other arms). Such items are trivially “grounded” and “blind-answerable”, and the judge still gave them 5 of 5 for correctness. The verifier’s feedback loop seems to push the small model towards copying the source verbatim, which is the cheapest way to pass a grounding check. The verifier now has two rule checks for this (the key must not be contained in the stem; the stem must be a question; Table II), and the leakage rate is reported as a metric.

Chunking. Fixed 800-character windows (E5c) were the weakest single-call variant: 45 % of question pairs were near-duplicates (10–16 % with semantic chunking) and the judge’s distractor score fell to 3.0 of 5 (3.5–3.9 with semantic chunking).

VIII. DISCUSSION

Why a method rather than a prompt. The pilot supports the design argument on three points: explicit planning improved grounding, blind answerability, distractor validity and coverage of the same 7B model; a much larger model given the document still concentrated on one part of it; and per-question citations let a reader—the judge here, the teacher in

TABLE IX

TWO PILOT QUESTIONS, TRANSLATED FROM HUNGARIAN (* = MARKED KEY).

E0	<i>In which century did [coffee] reach [Europe] through Venetian merchants? 9th*/10th/8th/11th century. Judge: not grounded—the source says the 17th century; the 9th is the century of the Ethiopian legend. No option is correct.</i>
E3	<i>“The habit of drinking coffee spread from Ethiopia to Yemen, then to the Arab world, and reached Europe in the 17th century through Venetian merchants.” Key: the same sentence*; distractors: the sentence with the 16th, 18th or 15th century. Judge: grounded, blind answer correct, correctness 5 of 5—although the stem gives the answer away.</i>

practice—check each key against its source, which exposed both the naive baseline’s wrong keys and the verified arms’ leakage. It also shows the limits: verification multiplies calls by 60–100, and a verifier that is the same 7B model adds cost without yet adding information. The component that must earn its place in the main study is therefore the verifier, not the blueprint.

From pilot to main study. Each pilot finding changes the design of the `eval-v1` run on the 39 documents (items (i)–(iv) are implemented and tested in the harness): (i) a *whole-document* baseline—the document in one prompt, locally and on the server—tests P1 and P2 directly instead of through naive RAG; (ii) the last-resort path keeps only well-formed questions; (iii) two rule checks reject questions whose stem contains the key or is not a question, and the leakage rate is reported; (iv) the verifier is evaluated as a classifier against the judge and the teachers, and a verifier from a different model family is compared with the self-verifier; (v) three teachers rate a blind sample so that conclusions do not rest on the LLM judge alone; and (vi) every arm is run on all documents, so that differences of the size the pilot shows can be separated from the 0.07 run-to-run noise. Its results will replace Table VI: [quality index per arm with 95 % CIs; paired tests for E1–E0, E2–E1, E3–E2, E4–E0; leakage rates; verifier and judge agreement with teachers].

Reproducibility. Every run stores its resolved configuration, git commit, dataset version and generator settings; judge answers are cached by a hash of model, prompt version and prompt, so re-scoring is deterministic; runs resume after interruptions; and the full run→score→report flow is covered by end-to-end tests with fake services and a mock LLM. The report also writes the paper’s tables (Tables VI and VII). The pilot’s raw exams, judge verdicts and report are released with the code.

Limitations. The pilot covers one short Hungarian document on which naive RAG sees three of four chunks, the setting least favourable to planning; long documents should show larger differences. The judge sees each question’s cited context and may miss contradictions elsewhere. Only multiple-choice questions are evaluated. The Hungarian and English subsets differ in source, so language effects are confounded with source.

IX. CONCLUSION

MIMIR turns exam generation from a single prompt into a verifiable method that runs on a consumer GPU without retaining teaching material: a blueprint guarantees the requested exam and spreads it over the document, every question cites its source and is checked before a teacher approves it, and a 7B model fits in 6GB of GPU memory. A pilot shows that planning improves a small model’s grounding, distractors and coverage, that a large model given the document still clusters its questions, that a same-model verifier and an LLM judge can both be fooled by questions that copy their answer into the stem, and that single-document comparisons are dominated by run-to-run noise. The bilingual 39-document study with teacher ratings tests these findings; code, configurations and dataset manifests are released for reproduction.

ETHICAL CONSIDERATIONS

Uploaded documents are deleted after each job and no student data is processed. Every generated question requires teacher review, and exports disclose AI assistance and the model used, in line with the EU AI Act [21]. Evaluation documents are the authors’ own or openly licensed (CC BY 4.0, CC BY-SA 4.0).

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